

Baseball Analytics & Player Performance Prediction

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Tools: Python · SQL · SQLite · scikit-learn · XGBoost · SHAP

Project Overview

This project applies industry-level data science techniques to the Lahman Baseball Database, a comprehensive historical record of Major League Baseball statistics dating back to 1871. The project demonstrates a full end-to-end analytics pipeline including database design, SQL querying, data cleaning, exploratory data analysis, and machine learning modeling.

Two predictive models were developed:

- 1. OPS Prediction** — five models were trained and compared including Linear Regression, Ridge, Lasso, Random Forest, and XGBoost. Ridge Regression performed best with an RMSE of 0.079 and R^2 of 0.432.
- 2. Hall of Fame Prediction** — Logistic Regression and XGBoost Classifier were compared. Logistic Regression achieved the best AUC-ROC of 0.896 with an 87% recall on inducted players.

Data & Methods

Data Source

The Lahman Baseball Database is a comprehensive historical record of Major League Baseball statistics from 1871 to 2025, maintained by Sean Lahman and the Chadwick Baseball Bureau. The database contains 28 tables covering batting, pitching, fielding, salaries, awards, Hall of Fame voting, and biographical information for over 24,000 players.

Technical Stack

SQLite was used to store and query the raw database tables. Python was used for all data cleaning, feature engineering, visualization, and modeling. Key libraries include pandas and numpy for data manipulation, matplotlib and seaborn for visualization, scikit-learn for machine learning pipelines, XGBoost for gradient boosting models, and SHAP for model explainability.

Project Structure

The project is organized into four phases: (1) SQL database design and exploratory queries, (2) Python data cleaning and feature engineering, (3) exploratory data analysis and visualization, and (4) machine learning modeling. All code is version controlled on GitHub.

SQL & Database Design

The Lahman CSV files were ingested into a SQLite database using Python's sqlite3 library. Six analytical views were created using industry-standard SQL techniques including CTEs, window functions, and multi-table joins. Two key examples are shown below.

Year-Over-Year Home Run Change

This view uses the LAG() window function to compare each player's home run total to their previous season, partitioned by playerID to ensure accurate year-over-year comparisons.

```
CREATE VIEW IF NOT EXISTS v_hr_yoy_change AS
WITH hr_stats AS (
    SELECT yearID, playerID, AB,
           SUM(HR) AS total_hr,
           LAG(SUM(HR)) OVER (PARTITION BY playerID ORDER BY yearID) AS lag_hr
    FROM batting
```

```

WHERE AB >= 300
GROUP BY playerID, yearID
)
SELECT p.nameFirst || ' ' || p.nameLast AS player_name,
       h.yearID, h.total_hr, h.lag_hr,
       h.total_hr - h.lag_hr AS lag_diff
FROM people p
JOIN hr_stats h ON p.playerID = h.playerID
WHERE h.yearID >= 2010
ORDER BY lag_diff DESC;

```

Primary Position

This view identifies each player's primary defensive position by finding the position with the most games played using a RANK() window function partitioned by player.

```

CREATE VIEW IF NOT EXISTS v_primary_position AS
WITH pos_counts AS (
    SELECT playerID, POS,
           SUM(G) AS total_games,
           RANK() OVER (PARTITION BY playerID ORDER BY SUM(G) DESC) AS pos_rank
    FROM fielding
    GROUP BY playerID, POS
)
SELECT playerID, POS AS primary_position
FROM pos_counts
WHERE pos_rank = 1;

```

Data Cleaning & Feature Engineering

The raw Lahman tables required significant cleaning before modeling. Key decisions are outlined below.

Era Filtering

Data was filtered to 1990 and beyond to ensure consistency in tracked statistics. Many pre-1990 columns such as intentional walks, sacrifice flies, and caught stealing contained large numbers of nulls due to historical record keeping limitations rather than true missing values.

Minimum At Bat Threshold

Players with fewer than 300 at bats in a season were excluded from the OPS prediction model. This filters out pitchers, bench players, and partial seasons which would introduce noise and extreme outliers into the target variable distribution.

Feature Engineering

Several features were engineered from the raw counting stats including On Base Percentage (OBP), Slugging Percentage (SLG), and OPS. Player age was calculated from birth year and season year. The target variable next OPS was created using pandas shift(-1) grouped by playerID to look forward one season per player.

Hall of Fame Dataset

A separate dataset was constructed for HOF prediction by aggregating career level stats from batting, fielding, awards, All Star appearances, and World Series wins. Active players were excluded by filtering on final game

date. Primary pitchers were removed since the model focuses on position players. Players with fewer than 1000 career at bats were excluded to remove cup-of-coffee players.

Exploratory Data Analysis

Exploratory analysis was conducted on the cleaned dataset to understand distributions, relationships between features, and key baseball insights before modeling.

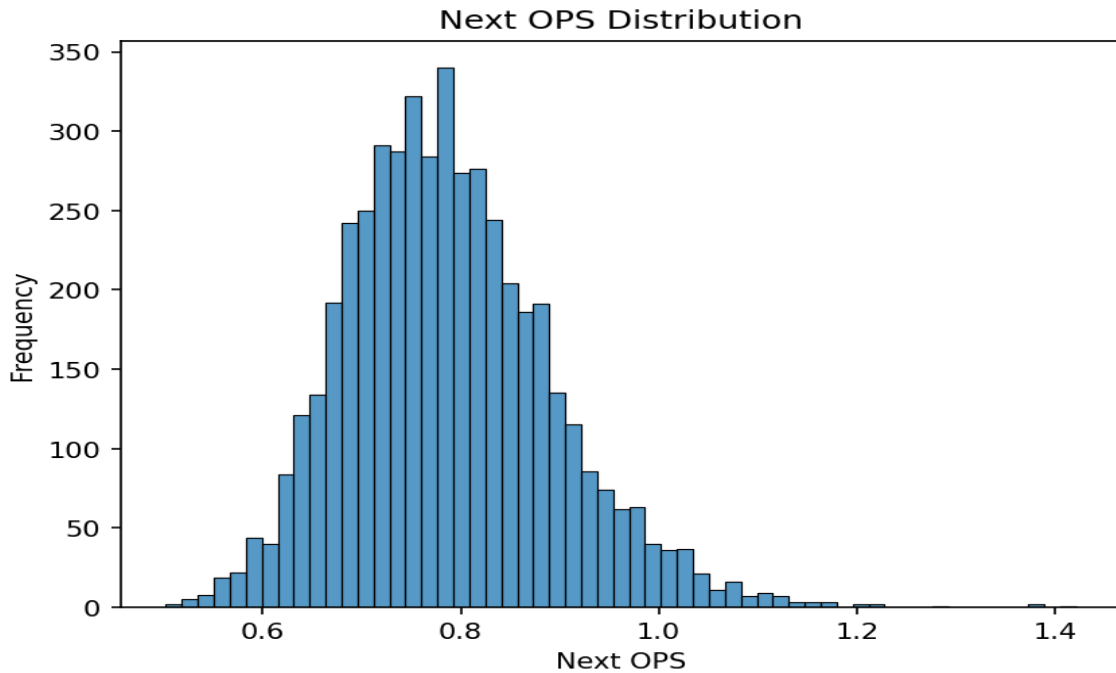


Figure 1: Distribution of next season OPS — roughly normal, ideal for regression modeling.

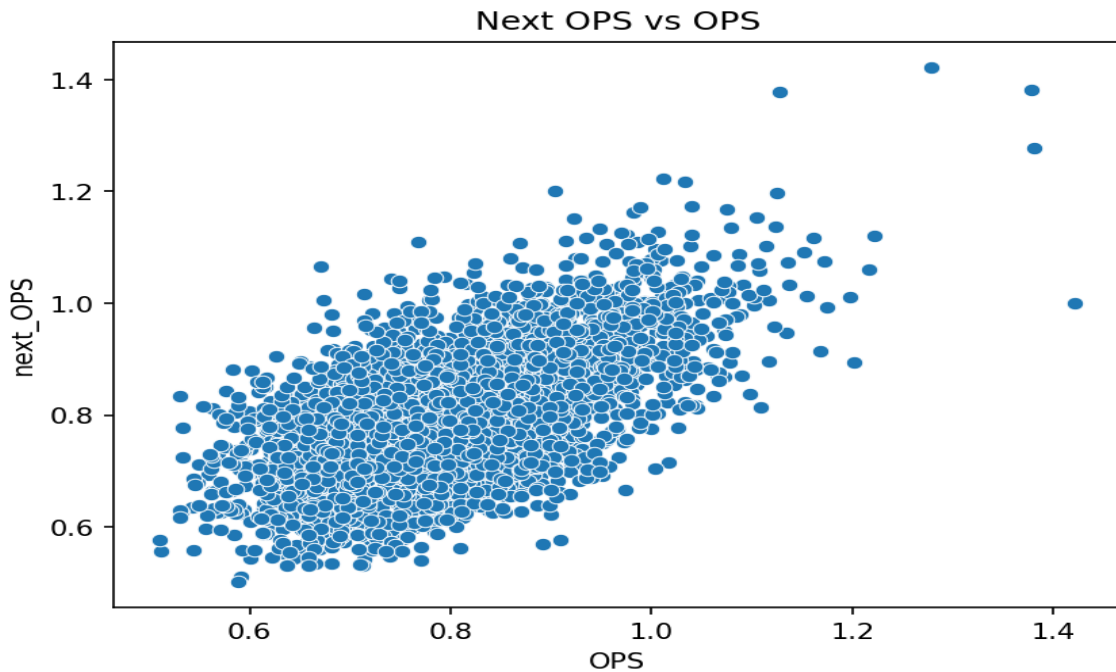


Figure 2: Next season OPS vs current OPS — strong linear relationship confirms OPS as the primary predictor.

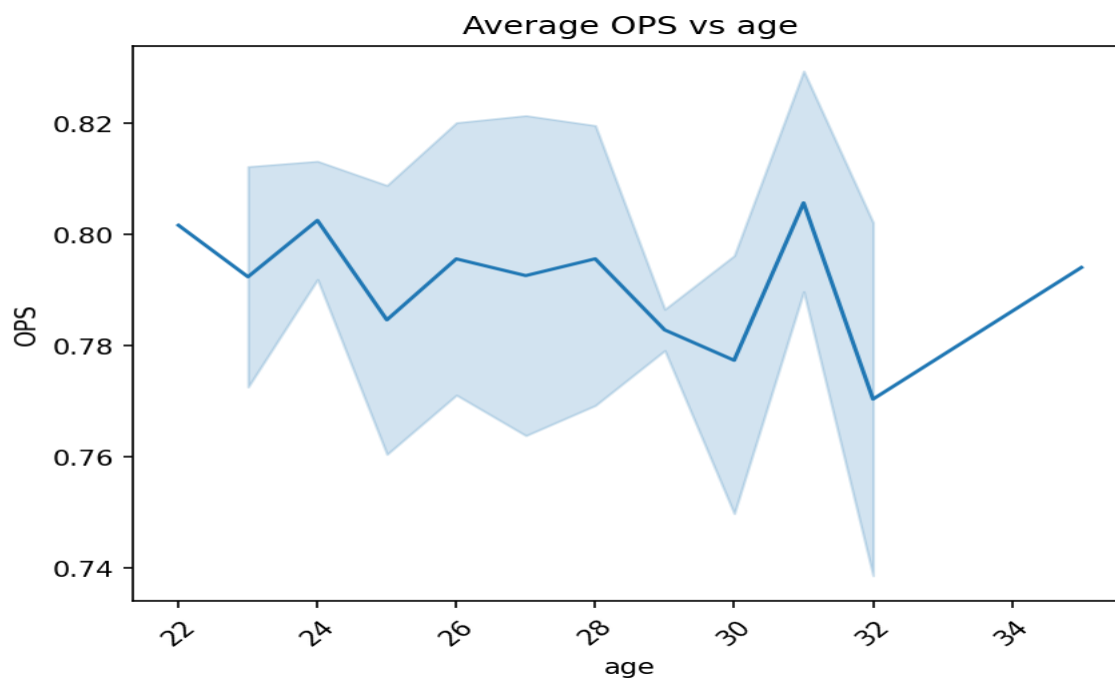


Figure 3: Average OPS by age — players peak offensively around age 28-31.

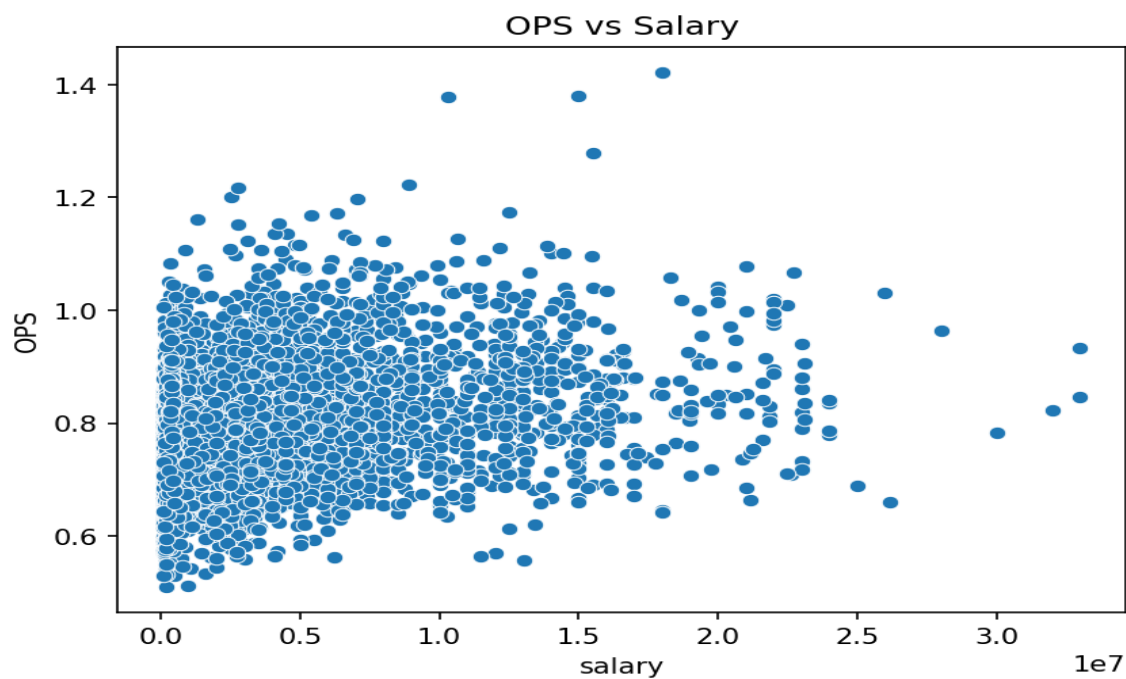


Figure 4: OPS Vs Salary — weak correlation suggesting teams don't always get what they pay for.

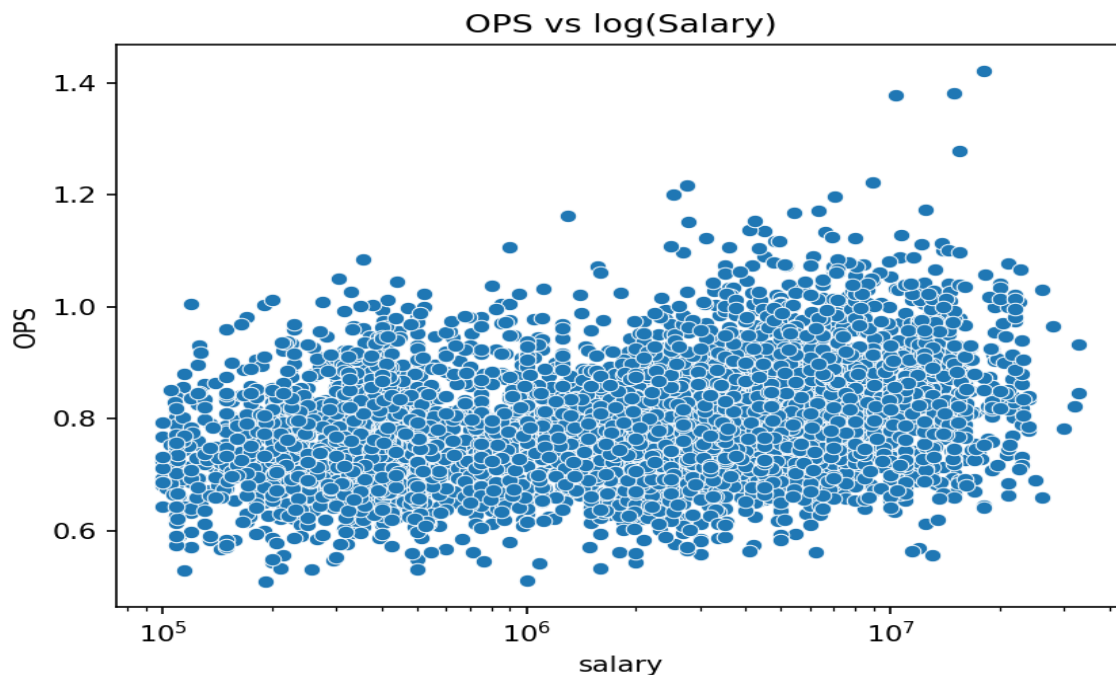


Figure 5: OPS vs Salary (log scale) — slight positive trend becomes visible with log transformation.

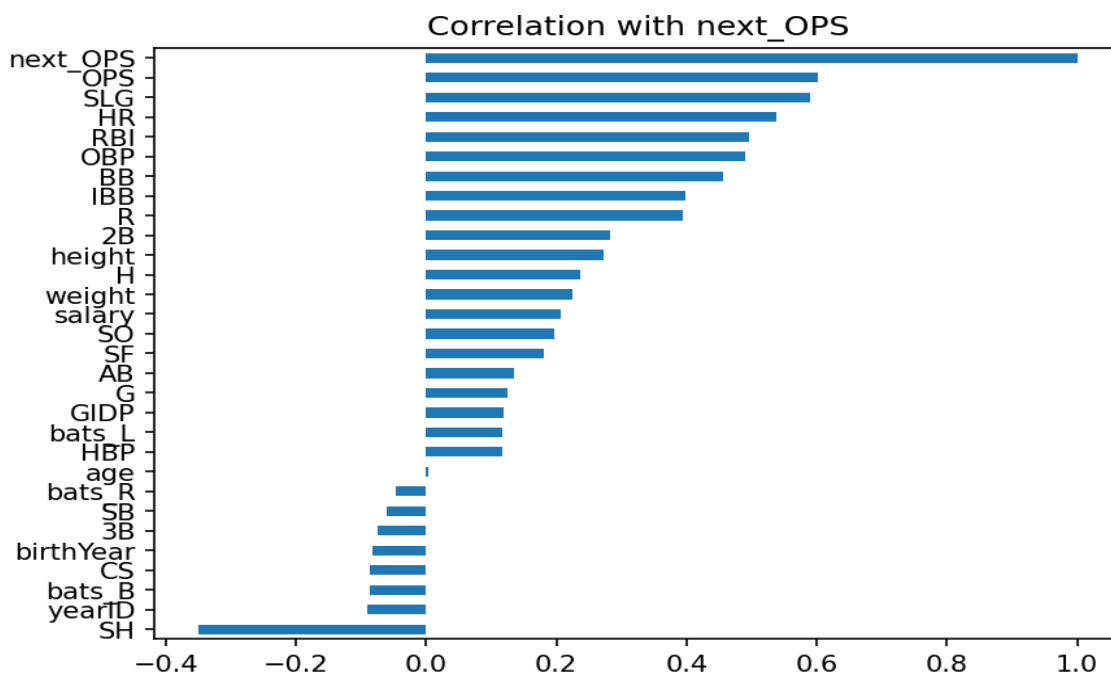


Figure 6: Feature correlations with next season OPS — OPS, SLG, and BB are the strongest predictors.

OPS Prediction Model

A regression model was built to predict a player's OPS in the following season using their current season statistics, age, and salary. Five models were trained and compared using 80/20 train/test split with cross validation for hyperparameter tuning.

Model Comparison

Model	RMSE	R ² Score
Ridge Regression	0.0786	0.432
Linear Regression	0.0786	0.431
Random Forest	0.0793	0.421
XGBoost	0.0796	0.417
Lasso Regression	0.0819	0.383

Ridge Regression performed best with an RMSE of 0.079 — meaning predictions are within roughly 0.079 OPS points on average. The dominance of linear models suggests that next season OPS has a largely linear relationship with current season statistics, limiting the advantage of complex ensemble methods.

SHAP Feature Importance

SHAP values were calculated for the Ridge model to identify which features most influenced predictions. Current OPS, SLG, and BB were the strongest predictors — confirming that plate discipline and power hitting are the most consistent and repeatable offensive skills.

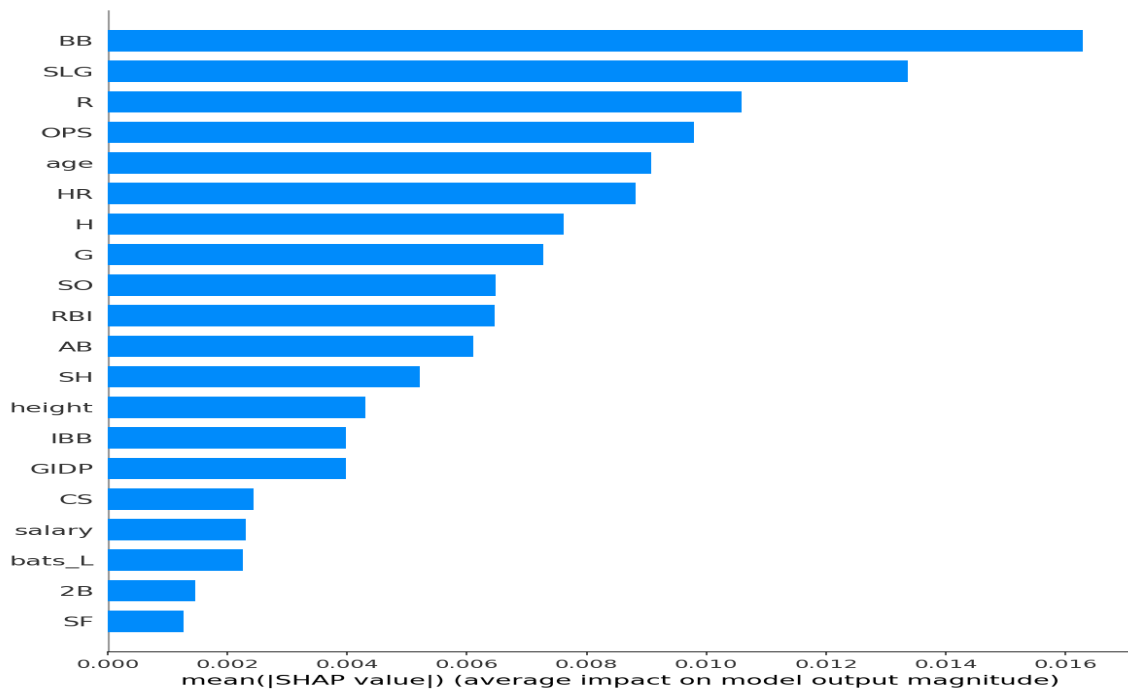


Figure 7: SHAP feature importance for OPS prediction — OPS, SLG, and BB dominate.

Hall of Fame Prediction Model

A classification model was built to predict the probability of a player being inducted into the Baseball Hall of Fame based on their career statistics. Features included career batting stats, fielding percentage, primary position, All Star appearances, awards, World Series wins, career longevity, and age at retirement. Active

players and primary pitchers were excluded from the dataset.

Class Imbalance

Only 5.1% of qualifying players in the dataset were inducted into the Hall of Fame. This severe class imbalance was handled using `class_weight='balanced'` in Logistic Regression and `scale_pos_weight` in XGBoost. AUC-ROC and recall were used as primary evaluation metrics rather than accuracy, which is misleading under class imbalance.

Model Comparison

Model	AUC-ROC	Recall (Inducted)	Precision (Inducted)
Logistic Regression	0.896	0.87	0.35
XGBoost Classifier	0.859	0.74	0.59

Logistic Regression outperformed XGBoost with a higher AUC-ROC of 0.896 and recall of 0.87. For Hall of Fame scouting purposes recall is the more important metric — it is better to flag too many candidates than to miss a deserving player. XGBoost showed higher precision but at the cost of missing more true inductees.

SHAP Feature Importance

Career seasons played, hits, RBI, and OBP were the strongest predictors of HOF induction. Notably home runs did not rank as the top feature — suggesting that longevity and consistency matter more to HOF voters than peak power. This finding aligns with the historical HOF voting patterns where players like Barry Bonds have been kept out despite elite statistics due to non-statistical considerations.

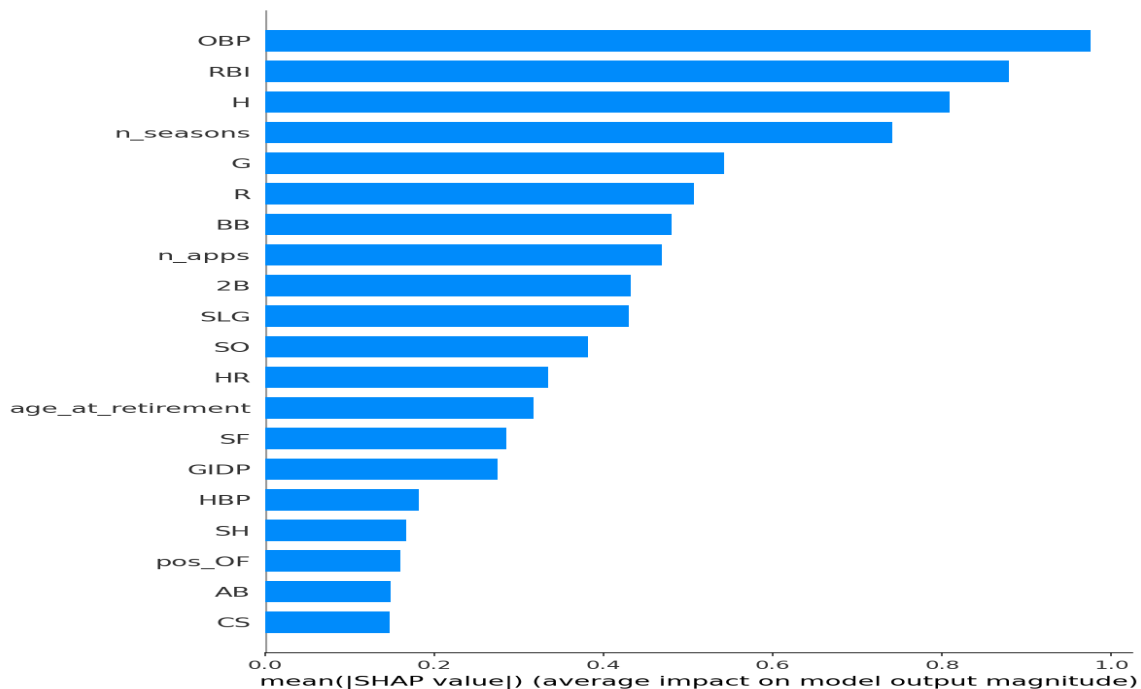


Figure 8: SHAP feature importance for HOF prediction — longevity and contact dominate over power.

Player Projections

Two prediction functions were built to make the models interactive. Given a playerID, each function queries the dataset and returns a prediction.

OPS Prediction Function

`get_next_ops_prediction(playerID)` — returns the predicted next season OPS for a given player based on their most recent season statistics.

```
get_next_ops_prediction('troutmi01') → Predicted OPS: 0.923
get_next_ops_prediction('judgear01') → Predicted OPS: 0.910
get_next_ops_prediction('bettsmo01') → Predicted OPS: 0.881
```

Hall of Fame Prediction Function

`make_hof(playerID)` — returns the probability of a player being inducted into the Hall of Fame based on their career statistics.

```
make_hof('bondsba01') → HOF Chance: 99.9%
make_hof('troutmi01') → HOF Chance: 95.0%
make_hof('abreubo01') → HOF Chance: 24.8%
make_hof('jeterde01') → HOF Chance: 99.9%
```

Conclusions & Key Insights

OPS Prediction

Current season OPS is the strongest predictor of next season OPS, confirming that offensive performance is highly consistent year over year for qualified hitters. Ridge Regression outperformed Random Forest and XGBoost, suggesting the relationship between features and next season OPS is largely linear. Walk rate (BB) ranked as a top SHAP feature, supporting the sabermetric finding that plate discipline is one of the most repeatable offensive skills.

Salary & Performance

Salary showed only a weak correlation with OPS, even on a log scale. This supports the Moneyball hypothesis that market inefficiencies exist in baseball — teams frequently overpay for declining veterans while undervaluing young players still under team control.

Hall of Fame Prediction

Career longevity as measured by seasons played was the most important HOF predictor, followed by career hits and RBI. This suggests HOF voters reward sustained excellence over peak performance. The model predicted Barry Bonds as a near-certain HOF candidate based purely on statistics, highlighting the gap between statistical merit and the actual voting process which incorporates character considerations.

Linear vs Complex Models

Linear models outperformed ensemble methods in both modeling tasks. This is consistent with the nature of aggregated baseball statistics — engineered features like OPS already capture most of the signal, leaving little non-linear structure for tree based models to exploit. This finding reinforces that model selection should be driven by the data rather than defaulting to the most complex available method.

Technical Stack

Tool	Purpose
SQLite	Database storage and querying
Python	Data cleaning, modeling, and reporting
pandas / numpy	Data manipulation and feature engineering
matplotlib / seaborn	Data visualization
scikit-learn	Machine learning pipeline and evaluation
XGBoost	Gradient boosting models
SHAP	Model explainability
reportlab	PDF report generation
GitHub	Version control and portfolio hosting